

Total No. of Questions : 4]

SEAT No. :

PD-321

[Total No. of Pages : 2

[6411]-100

B.E. (Mechanical) (Insem.)

ENERGY ENGINEERING

(2019 Pattern) (Semester - VIII) (402049)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Attempt Q.1 OR Q.2, Q.3 OR Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right of each question indicate full marks.
- 4) Assume suitable data wherever necessary and mention the same clearly.
- 5) Use of steam tables, Mollier chart and calculator is allowed.

- Q1) a) In a two stage turbine working on ideal Rankine cycle, steam enters the first stage at 90 bar and 550 °C temperature. Second stage entry temperature after reheating is 550 °C. Steam is never wet during expansion in both, sides. Find the cycle efficiency and specific steam consumption. Also find the steam flow rate in the turbine in kg/s if the turbine is directly connected to an alternator of 150 MW output. Assume 0.9 as alternate efficiency. Draw T-S diagram for cycle. Neglect pump work. [6]
- b) Explain the Fluidized Bed Combustion (FBC) boiler and Circulating FBC system. [5]
- c) Write a note on : Present Status of Power Generation in India. [4]

OR

- Q2) a) In a single-heater regenerative cycle the steam enters the turbine at 30 bar, 400°C and the exhaust pressure is 0.10 bar. The feed water heater is a direct contact type which operates at 5 bar. Determine the following,
- i) The cycle efficiency and the steam rate of the cycle.
 - ii) The increase in mean temperature heat addition, efficiency and steam rate as compared to the Rankine cycle (without regeneration).
 - iii) Draw flow diagram, T-s and h-s diagrams also.
- Note: Pump work may be neglected. [6]
- b) Explain with neat sketch 'Lamont Boiler'. [5]
- c) Estimate the available energy sources in Maharashtra. [4]

P.T.O.

Q3) a) The following observations were recorded during a trial on a steam condenser. [6]

Vacuum = 700 mm of Hg

Barometer reading = 76.5 cm of Hg

Mean condenser temperature = 35 °C

Hot well temperature = 28 °C

Condensate collected = 1800 kg/hr.

Air leakage = 1 kg/ton of steam

Cooling water circulated = 70 ton/hr.

Cooling water inlet temperature = 12 °C

Cooling water outlet temperature = 27 °C

Calculate;

- i) Vacuum efficiency
- ii) Condenser efficiency
- b) Write detailed note on carbon credits and its implications on clean technology. [5]
- c) Write a note on 'Next Generation Cooling Towers'. [4]

OR

Q4) a) In a surface condenser a section of the tubes near to the air pump suction is screened off so that the air is cooled to a temperature below that of the condensate, separate extraction pumps being provided to deal with air and condensate respectively. 5448 kg of steam are condensed per hour and the air leakage is 4.54 kg/hr. The temperature of the exhaust steam is 31°C, the temperature of the condensate is 27°C, and the temperature at the air pump suction is 21.1°C. Assuming a constant vacuum throughout the condenser, determine the following

- i) The mass of the steam condensed per hour in the air cooler;
- ii) The volume of air in m³/hr to be dealt with by air pump;
- iii) The percentage reduction in necessary air pump capacity following the cooling of the air. [6]
- b) Classify cooling towers. Explain with neat sketch 'Induced Draught Cooling Towers'. [5]
- c) Discuss various pollutants from thermal energy based power plants. [4]

